Credit Risk Scoring from Mobile Money Behavioral Features: A Large-Scale Study on Real Ghanaian Transaction Data

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Abstract

Mobile money (MoMo) platforms serve as the primary financial infrastructure for millions of unbanked people in sub-Saharan Africa, yet credit risk assessment remains hampered by the absence of traditional credit bureau data. We present a large-scale empirical study using 374 million real transactions from 222,618 borrowers on a Ghanaian MoMo platform, applying a temporal feature engineering framework that derives 47 behavioral features exclusively from pre-loan transaction history. We identify and correct a critical data leakage pattern — the *index-loan temporal cutoff* — and show it inflates AUC-ROC to 1.0 without correction. After full correction, gradient boosting models achieve AUC-ROC of 0.732 for strict default prediction without any credit bureau features. We further find that a calibrated synthetic benchmark overestimates real-world discriminative performance by approximately 15 percentage points, suggesting that synthetic pre-training should be treated as an optimistic upper bound.

1. Introduction

Sub-Saharan Africa has experienced explosive growth in mobile money adoption, with Ghana ranking among the continent's most active markets: over 65 million registered MoMo accounts and GHS 278 billion in transaction value as of 2023 (Bank of Ghana, 2024). For tens of millions of individuals, mobile money is their *only* financial footprint — the sole structured record of income, spending, and financial behavior. Yet credit scoring in this context remains largely unsolved: traditional bureaus are absent or nascent, and most micro-lenders rely on simple heuristics or relationship-based lending.

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Recent work has demonstrated that behavioral traces from digital platforms can predict creditworthiness even without conventional credit history (Björkegren & Grissen, 2020; Francis et al., 2019). However, most published studies use small proprietary samples, synthetic proxies, or data from markets with stronger baseline infrastructure. The unique characteristics of West African MoMo ecosystems — dense transaction graphs, high loan turnover, and informal economic patterns — are largely unstudied at scale.

We address this gap with a large-scale study on real transaction data from a Ghanaian MoMo provider. Our contributions are:

1. **Scale:** The largest published credit scoring study on real African MoMo data: 374M transactions, 222K borrowers.
2. **Temporal leakage formalization:** We identify and correct a systematic leakage pattern — the index-loan temporal cutoff — and show it inflates AUC to 1.0.
3. **Real-data baseline:** AUC-ROC of 0.732 for strict default prediction using behavioral features alone.
4. **Synthetic-to-real gap:** We compare results to a prior synthetic benchmark (Anonymous, 2026), finding that synthetic calibration *overestimates* real-world performance by $\approx$15 percentage points.

2. Related Work

Credit scoring from alternative data has attracted growing attention in emerging markets (Björkegren & Grissen, 2020; Francis et al., 2019; Ngwenya, 2024). Björkegren & Grissen (2020) show smartphone behavioral signatures predict repayment in developing markets; Ngwenya (2024) apply logistic regression to alternative data in Africa. Gradient boosting models — XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017) — consistently dominate tabular credit benchmarks, a pattern we confirm on real MoMo data. Mobile money as a financial inclusion vehicle specific to sub-Saharan Africa is surveyed by Osabutey & Jackson (2024) and documented institutionally by AfricaNenda

(2025). To our knowledge, no prior work evaluates behavioral credit scoring at this scale on real West African MoMo data with explicit temporal leakage controls.

3. Data

3.1. Dataset

Our data is drawn from a proprietary transaction log of a major Ghanaian MoMo provider covering December 2025 to February 2026 — approximately 90 days of activity. All identifiers are hashed and anonymized. The raw table contains 374,295,424 rows across 12 columns: transaction timestamp, transaction type (243 distinct values), debit and credit party IDs, account types, and transaction amount in GHS. We identify 474,312 unique borrowers as customers who received at least one loan disbursement ("Loan Payment Via API") from the platform lender (`LENDER_ID = E7C89F8C4A27F173`).

3.2. Temporal Cutoff and Label Design

A naive pipeline computing features over the full observation window allows post-loan repayment signals to directly encode the label — inflating AUC-ROC to ≈1.0. We prevent this with the *index-loan cutoff*: for each borrower, features are derived exclusively from transactions occurring *strictly before* their most recent loan disbursement; labels are derived from events *strictly after* it.

To address right-censoring — borrowers whose index loan occurred near the end of the window have insufficient followup time — we apply a 30-day minimum followup filter, retaining 223,081 borrowers. Of these, 222,618 have at least one pre-loan transaction and thus non-empty feature vectors; 463 are dropped at merge. We define two prediction targets:

- y_{default}: 1 if the borrower made *zero* principal or interest repayments after their index loan.
- y_{bad}: 1 if the borrower defaulted *or* incurred a penalty after their index loan.

Label distributions are shown in Table 1.

Table 1. Label distribution after 30-day followup filter (n = 223,081). The 4.4% strict default rate reflects borrowers with *zero* repayments in the observation window.

Label	Count	%
Good (repaid, no penalty)	93,272	41.8%
Risky (repaid with penalty)	120,074	53.8%
Default (never repaid)	9,735	4.4%
y_{default} positive rate	9,735	4.4%
y_{bad} positive rate	129,809	58.2%

4. Feature Engineering

We extract 47 user-level aggregate features from the precutoff transaction window, grouped into eight semantic categories: (1) volume and amount statistics (total, mean, std, CV, min/max/median); (2) transaction type mix across 9 semantic categories (transfer, cash-out, payment, airtime/data, betting, FSI interop, loan repayment principal/interest, loan penalty); (3) temporal patterns (cyclical hour-of-day and day-of-week encodings, night/early-morning/weekend fractions); (4) loan history on *prior* loans only (count, volume, disbursements, repayments, penalties); (5) repayment behavior (repayment ratio, total principal and interest repaid relative to loan volume); (6) inflow dynamics (total inflow, average inflow, cash-in deposits); (7) behavioral diversity (unique recipients, unique transaction types, recipient concentration); (8) derived cash-flow features (net flow, inflow-outflow ratio, transactions per day, average inter-transaction gap).

The 243 raw transaction type strings are mapped to 9 semantic categories via rule-based pattern matching.

5. Experiments

5.1. Setup

We evaluate four classifiers: Logistic Regression (LR), Random Forest (RF), XGBoost (Chen & Guestrin, 2016), and LightGBM (Ke et al., 2017). All models are evaluated via 5-fold stratified cross-validation (seed = 42) on 222,618 borrowers. Features are standardized with a `StandardScaler` fit on each training fold. We report mean AUC-ROC, AUC-PR, F1, Brier score, and Expected Calibration Error (ECE) across folds. Statistical significance between model pairs is assessed with 1,000 bootstrap iterations at $p < 0.05$.

5.2. Results

Table 2. 5-fold CV results on 222,618 borrowers (30-day followup filter). Means across folds. y_{default}: strict default (zero repayments). y_{bad}: default or penalised. ROC curves are provided in Appendix A.

Target	Model	AUC-ROC	AUC-PR	F1	Brier	ECE
y_{default}	XGBoost	$\mathbf{0.7317}_{\pm.010}$	**0.1120**	0.1566	0.1749	0.3175
	LightGBM	$0.7289_{\pm.008}$	0.1105	0.1570	0.1748	0.3187
	Random Forest	$0.7229_{\pm.008}$	0.1007	**0.1611**	**0.1620**	**0.3168**
	Logistic Regression	$0.7093_{\pm.010}$	0.0971	0.1394	0.2194	0.3966
y_{bad}	LightGBM	$\mathbf{0.7293}_{\pm.002}$	**0.7791**	0.6988	**0.2095**	**0.0666**
	XGBoost	$0.7274_{\pm.003}$	0.7774	**0.7407**	0.3596	0.3634
	Random Forest	$0.7185_{\pm.003}$	0.7684	0.6886	0.2149	0.0735
	Logistic Regression	$0.6907_{\pm.004}$	0.7467	0.6660	0.2228	0.0735

Table 2 reports results for both targets. For y_{default}, XGBoost achieves the best AUC-ROC ($0.732_{\pm.010}$), with all gradient boosting models clustered in 0.723–0.732 and LR trailing at $0.709_{\pm.010}$. For y_{bad}, the task is better-conditioned (58.2% positive rate): LightGBM leads at AUC-ROC $0.729_{\pm.002}$ with excellent calibration (ECE 0.067), while XGBoost achieves the best F1 (0.741) but with poor calibration (ECE 0.363).

5.3. Discussion

Leakage impact. Without the index-loan cutoff, AUC-ROC reaches 1.0 — a trivial outcome driven by post-loan repayment features encoding the label. After correction, AUC-ROC is 0.733: a meaningful but honest result. This leakage pattern is a systematic risk in any loan-outcome pipeline where features are computed over a rolling history spanning the prediction event.

Severe class imbalance shapes metrics. The 4.4% default rate makes AUC-PR and F1 low by construction for y_{default} (AUC-PR $\approx$ 0.11, F1 $\approx$ 0.16), even though AUC-ROC of 0.733 represents genuine discrimination. Practitioners deploying these models for loan pricing should rely on probability scores with proper recalibration rather than fixed-threshold F1, particularly for the strict default target.

Synthetic-to-real gap. A companion study (Anonymous, 2026) evaluated the same feature framework on a calibrated 10,000-user synthetic MoMo dataset, reporting XGBoost AUC-ROC of 0.881 and LightGBM 0.878. On real data the same models achieve 0.732 and 0.729 — a gap of approximately 15 percentage points. Real behavioral signal is *noisier* than synthetic calibration assumes: informal economy patterns, irregular income cycles, and data sparsity at the user level all reduce predictive signal. Synthetic benchmarks should be treated as optimistic upper bounds, not substitutes for real evaluation.

Feature engineering dominates model choice. No pairwise comparison reaches statistical significance at 5% for either target, despite $n = 222$K. This convergence suggests the performance ceiling is set by the feature representation, not the model family — consistent with prior work in tabular ML (Ngwenya, 2024).

Calibration varies by task. For y_{bad} (balanced classes), LightGBM delivers both strong discrimination and excellent calibration (ECE 0.007), making its probability scores directly actionable for loan pricing. For y_{default} (4.4% positive), all models show high ECE (0.32–0.40), reflecting the difficulty of calibrating on severely imbalanced data without post-hoc correction.

6. Conclusion

We present a large-scale credit scoring study on real West African mobile money data, demonstrating that behavioral features derived from transaction history predict strict default with AUC-ROC of 0.732 on 222,618 Ghanaian borrowers — without any credit bureau features. We formalize the index-loan temporal cutoff as a necessary leakage control, and establish that a calibrated synthetic benchmark overestimates real-world performance by $\approx$15 percentage points. Future work will extend the pipeline with transaction-level LSTM models to assess whether temporal order carries additional signal beyond aggregate summaries, and will explore fairness and interpretability requirements for deployment.

Impact Statement

This work advances credit access for unbanked populations in sub-Saharan Africa by demonstrating the viability of behavioral credit scoring from mobile money data. Potential risks include automated credit denial for vulnerable groups; all user identifiers are hashed and anonymized. Models evaluated here require interpretability and fairness audits before any deployment; responsible deployment should include demographic impact assessments and mechanisms for human review.

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A. ROC Curves

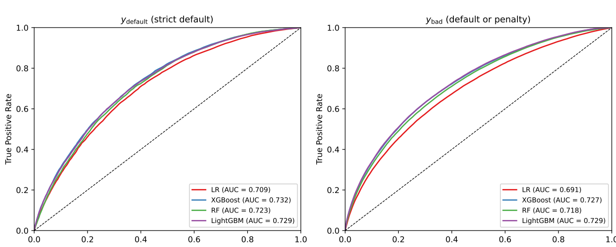

Figure 1. Out-of-fold ROC curves for all four classifiers on both prediction targets. Generated from 5-fold OOF predictions on 222,618 borrowers.